

Table 5: Parameter Count and Training Mechanism Comparison
 $(n = 20, h = 256, \tilde{L} = 2, L = 4, p = 2)$

P1 DQN (tier)	Shinde & Tarchi [1]	48,128	$O(nh)$	SGD on TD error
P2 DQN (node)	Shinde & Tarchi [1]	48,384	$O(nh)$	SGD on TD error
P3 DQN (ratio)	Shinde & Tarchi [1]	47,616	$O(nh)$	SGD on TD error
Tri-DQN HRL [1]	Combined	144,128	$O(3nh)$	3× gradient descent
VQC (tier+ratio)	Quantum-HRL (ours)	20	$O(L\log_2 n)$	PSR + Bayesian Opt.
QAOA (node)	Quantum-HRL (ours)	4	$O(p)$	Bayesian Opt. on $C(\gamma, \beta)$
Quantum-HRL	Combined	24	$O(L\log_2 n + p)$	Bayesian Opt.
Reduction factor ρ	--	6,005 ×	--	--